Toni Guan, Ziyan Liu, Yingrui Gao, Chuhan Feng, Yue Li
College of Information Technology
Shenyang Institute of Technology
Shenyang, Liaoning, China
guantong490@gmail.com

October 15, 2026

Editor-in-Chief
Computer Methods and Programs in Biomedicine
Editorial Office

Dear Editor,

We are pleased to submit our original research article entitled "**When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study**" for your consideration as a **Full Length Article** in *Computer Methods and Programs in Biomedicine*. This manuscript has not been published elsewhere and is not under consideration by any other journal.

Summary of contributions. Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We pre-registered and executed a multi-seed calibration boundary study comprising **60 experiments** (6 directed transfer pairs $\times$ 2 architectures $\times$ 5 seeds, fully balanced) on PTB-XL, Chapman-Shaoxing, and CPSC2018, with a patient-level cluster paired bootstrap (the operating CI is BCa at $B = 10{,}000$, fully re-run on the entire 60-experiment grid) and eight recalibration methods. TS yields a positive OOD calibration benefit $\Delta\mathrm{ECE}_{\mathrm{OOD}} = \mathrm{ECE}_{raw}^{OOD} - \mathrm{ECE}_{TS}^{OOD}$ in **45/60 = 75.0%** of seed experiments (InceptionTime 24/30, ResNet1D 21/30), while the remaining **15 counter-examples are transparently disclosed** and mechanistically explained case by case (seven co-occur with low OOD accuracy and a large ID–OOD accuracy drop; the remainder are seed-level temperature mis-transfers, one of which is over-correction on an already well-calibrated backbone). The paper delivers **five independent contributions**: **C1** (Primary, OOD benefit quantification: 45/60 seed support), **C2** (Boundary condition, pre-registered branch triggered: only $30/60 = 50.0\%$ of cells have BCa 95% CIs containing zero, below the $\geq$80% criterion, so the "ID nonzero boundary" branch applies and C1 is reported with this qualifier), **C3** (Methodological, Shapley decomposition validation: 27/27 cells pass at $n = 20{,}000$), **C4** (Practical, method-selection boundary: TS the most consistent recalibrator *among the eight delivered methods* vs. EM prior, which is negative on average, and Matrix scaling, governed by a prior-likelihood mismatch mechanism), and **C5** (Clinical, discrimination-aware calibration gate: 5/12 pair$\times$architecture strata pass both layers under the per-seed majority criterion, reframing the counter-examples as fifteen Layer-1 rejections, seven of which also fail the discrimination layer).

Why this journal. Cross-corpus ECG transfer calibration is a critical bottleneck for clinical deployment of deep ECG models across institutions with different acquisition protocols, lead sets, and label vocabularies. Our contribution is evaluated end-to-end on the raw 12-lead ECG signal:

deep classifiers trained on one corpus are pushed through an eval-time shift ladder (sampling rate, lead availability, noise, gain) and a six-direction cross-corpus transfer grid, and the contribution is a boundary-condition characterization of when post-hoc recalibration pays off at deployment—combined with transparent counter-example reporting, a discrimination-aware deployment gate, and a pre-registered failure branch. We believe this falls squarely within the journal's scope on biomedical signal processing, reliability and calibration of clinical biosignal models, and reproducible evaluation methodology.

Transparency statement. All hypotheses in this manuscript are tagged *exploratory + robustness validation*, not confirmatory, consistent with the open-science best practices of Nosek et al. (2018). The fifteen counter-examples (25% of the experiment grid) are disclosed in full in the main post-hoc excluded. Four pre-registration amendments are recorded with reason and date: A1 re-locates the primary endpoint (protocol v2.0 $\rightarrow$ v2.1-A1), A2 reduces the confirmatory family from 156 to 12 hypotheses, re-designating the L2 shift levels as exploratory dose-response, A3 corrects the identity of the third corpus (the registered "CPSC2018+2019" is in fact *CPSC2018 only* – the CPSC Database plus the CPSC-Extra Database) and discloses that CPSC2018's official nine-class taxonomy contains no MI, so the MI class of our four-class subspace is populated entirely by CPSC-Extra SNOMED annotations, and A4 (2026-10-11) re-operationalizes the protocol's registered mapping-sensitivity gate on the delivered primary endpoint: the registered wording conditioned the gate on the secondary endpoint G, which its own Limitations acknowledge as undelivered on real data, so the gate now reads cross-pair $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ ordering (Kendall $\tau \geq 0.8$) with the confirmatory conclusions unchanged, defined *before* any mapping-variant result exists. A3 changes no recorded value (naming, attribution and disclosure); A4 changes no recorded value (gate operationalization only; the variant battery remains a queued deliverable). The protocol and the amendments are archived as an immutable record with per-file SHA-256 checksums at `https://doi.org/10.17605/OSF.IO/53H64` (amendments A3 and A4 are deposited in the OSF project deposit – project 53h64 Files, 2026-10-11, per-file SHA-256 verified; the frozen registration WJZ3Y remains intentionally at v1, the amendment set lives in the project deposit). A five-item honesty declaration (adversarial audit product) is included in the manuscript.

Data and code availability. The full pipeline code, the patient-level split indices, and the per-experiment result artifacts are already publicly available at `https://github.com/Toni-cs/ecg-calibration-transportability`, with trained model checkpoints for the 60-experiment grid provided as release assets of the same repository. The repository will be archived on Zenodo with a DOI upon acceptance. All three ECG corpora are public (PhysioNet/Kaggle).

This is a five-author submission; the authors share responsibility for all aspects of the work (see the CRediT statement in the manuscript). Highlights, the CRediT contribution statement, and a

data statement are provided with the manuscript. We look forward to your feedback.

Sincerely,

Toni Guan, Ziyan Liu, Yingrui Gao,
Chuhan Feng, Yue Li
College of Information Technology
Shenyang Institute of Technology
Shenyang, Liaoning, China
guantong490@gmail.com